Unit 3 Summary: Two-Way Tables

Name _____

Parts of a Two-Way Table

60 students answered two questions: *Do you play a sport? Do you have a job?* A two-way table shows the answers to both questions at once. Every example on this sheet uses this table.

	Job	**No job**	**Total**
Sport	8	22	30
No sport	12	18	30
Total	20	40	60

Part	**What it is**	**Example**
Categories	The answers to each question. They name the rows and the columns.	Rows: Sport, No sport. Columns: Job, No job.
Cell	Counts students who match **both** labels.	8 students play a sport **and** have a job.
Totals	In the margins: the **Total** row and the **Total** column.	30 students play a sport. 20 have a job.
Grand total	The corner.	60 students in all.
Check	Each row adds to its total. Each column adds to its total.	$8 + 22 = 30$. $8 + 12 = 20$.

Venn Diagram and Two-Way Table

Venn diagram	**Two-way table**	**Example**
Overlap: both groups	The cell for Sport **and** Job.	8 play a sport **and** have a job.
One circle only	The cell for Sport and No job. The cell for No sport and Job.	Sport only: $30 - 8 = 22$. Job only: $20 - 8 = 12$.
Outside the circles: neither group	The cell for No sport and No job.	$22 + 8 + 12 = 42$ are in at least one circle. Neither: $60 - 42 = 18$.
A **whole circle**	That group's total.	The "Plays a sport" circle holds $22 + 8 = 30$ students, not 22.

Percents from a Two-Way Table

	Job	No job	Total
Sport	8	22	30
No sport	12	18	30
Total	20	40	60

Kind	What it is	Example
Relative frequency table	Divide **every** count by the **grand total**. This gives the percent of **all** students.	$\frac{8}{60} \approx 13\%$. About 13% of **all** the students play a sport and have a job. The four inside cells add to 100%.
Marginal distribution	Percents from the **totals** (the margins). Divide by the **grand total**.	Job: $\frac{20}{60} \approx 33\%$. No job: $\frac{40}{60} \approx 67\%$.
Conditional distribution	Percents inside **one group**. Divide by that group's **total**.	Students who play a sport. Job: $\frac{8}{30} \approx 27\%$. No job: $\frac{22}{30} \approx 73\%$.

Check the denominator: "Of the students who **play a sport**, what percent have a job?" Divide by the sport total (30). Do not divide by 60. Do not divide by the job total (20).

Association

Step	What to do	Example
1. Find	Find the conditional distribution of each group.	Sport: $\frac{8}{30} \approx 27\%$ have a job. No sport: $\frac{12}{30} = 40\%$ have a job.
2. Compare	Compare the conditional distributions of the two groups. A segmented bar chart shows them side by side.	27% and 40% are **different**.
3. Decide	Conditional percents are **different** in the groups: the variables are **associated**. Conditional percents are about the **same**: there is **no association**.	Playing a sport and having a job are **associated**. In this group, students who play a sport are less likely to have a job.

Cause: an association only says the two things happen together. It does not prove cause.